

Benchmark for Crack Re-Identification in Ticket Management Systems

Délcio Ferramenta, Cátia Loureiro, António Silva, Beatriz Miranda, Paulo Brito, Nuno da Costa, Ricardo Rodrigues, and Du
DTx Colab, Campus de Azurém, Universidade do Minho, 4800-058 Braga, Portugal

Abstract—Building ticket management systems must determine whether a newly reported defect corresponds to an existing ticket or requires a new one. Repeated reports of the same crack from different viewpoints fragment history and duplicate work. We study this association as crack re-identification and introduce CRACKID (140 photographs, 17 walls, 212 identities in the frozen benchmark snapshot: 6 validation walls and 11 test walls with 19 matchable test identities). We evaluate ten zero-shot crop-based matchers from four families under retrieval, thresholded decisions, and open-set identification. Retrieval varies (Rank-1 0.352–0.654), but validation-calibrated pairwise decisions stay at or below the best-constant floor (F_1^{ov} 0.293–0.351 against a 0.305 floor; oracle F_1^* 0.305–0.359). Context and crack-erasure controls implicate wall location over crack appearance. A geometric baseline registering wall texture with cracks masked out and matching projected masks reaches 0.768 detection-and-identification rate at 0.1 false alarms on 48% of pairs, against 0.429 for the best crop method, 0.404 for the best context control, and 0.000 for a coverage-only control. All figures are point estimates on a small wall-clustered sample with single-annotator homography-propagated labels, so the open-set advantage is provenance-conditioned and ranking is short-baseline within-visit correspondence. This establishes a dataset and benchmark for the visual association step, leaving cross-visit performance and ticket reduction for future work.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Building ticket management systems organise reports of physical defects into records that support inspection, prioritisation, intervention, and follow-up. A ticket should maintain the history of one incident, but the same crack may be photographed repeatedly by different users or inspections, spawning duplicate tickets that fragment history and duplicate assessment work.

Addressing this requires associating each new observation with its existing ticket, or creating a new ticket for unrecorded damage.

Photographic evidence makes this a visual re-identification problem. Two cases matter: (i) unchanged morphology under varying viewpoint, distance, field of view, illumination and partial visibility; and (ii) physically evolved damage (extension, widening, branching) whose identity must persist despite condition change, distinguishing damage change from acquisition change. Either error creates a duplicate or misattributes the observation.

Existing crack methods address detection and segmentation within single images. Datasets such as CrackSeg9k [1] and OmniCrack30k [2] lack cross-observation identities, so detection cannot link an observation to a ticket. Whether person/object re-identification [3], [4] transfers to cracks—where distinct defects look alike and one defect varies widely—is untested.

We introduce CRACKID for deciding whether observations show the same wall crack: 140 photographs of 17 walls in a frozen snapshot (6 validation, 11 test; 19 matchable test identities, $n_Q=280$). Real photos cover acquisition variation within one visit; a synthetic protocol preliminarily probes appearance change and growth. Real cross-visit performance remains future work. All accuracy figures are point estimates on a small wall-clustered sample.

Our benchmark tests ten matchers zero-shot on CRACKID, separating candidate retrieval from thresholded same-crack decisions and open-set identification with rejection: ranking a record highly does not justify attaching a report to it. Context and crack-erased controls separate crack cues from wall location, and a geometric baseline registers walls and compares projected masks.

The study asks RQ1 (association under acquisition change), RQ2 (roles of context and geometric alignment), and RQ3 (behaviour under synthetic appearance/growth change). RQ1–RQ2 use real single-visit photos; RQ3 is exploratory synthetic. Ticket-level duplicate reduction and cross-visit evaluation are left for future work.

Contributions: (i) CRACKID with cross-observation identities and stated annotation provenance (Sec. III-B); (ii) a benchmark separating retrieval, same-defect decisions, and open-set identification (Sec. IV, VI); (iii) context controls and a geometric reference with coverage analysis (Sec. V, VI-C, VII-B); (iv) a preliminary synthetic revisit probe (Sec. VII-D).

II. RELATED WORK

Crack analysis targets detection and segmentation [5]–[7]; its datasets annotate single images and consolidations inherit that scope [1], [2], and multi-view pipelines fuse one inspection into a surface or segmentation [8] without instance correspondence. Structural-health monitoring compares cracks over time with related machinery [9]–[12], but identity is *given by construction* (fixed camera, marked location): they measure a known crack rather than deciding association under an open-set protocol, which is our target.

Object re-identification [4] with open-set identification [13] is our formal setting; keypoint matching [14], [15], learned correspondence (SuperGlue [16], LoFTR [17]), and region embeddings [3], [18]–[21] apply to crack crops, all benchmarked here zero-shot without CRACKID fine-tuning (YOLOv8 row: layer-9 `yolov8n.pt` features [22]; LightGlue [23] and NetVLAD [24] left as future retrieval baselines). Vehicle-damage re-identification across claims [25] mirrors our open-set protocol; the rest of that literature covers detection/severity rather than association [26]–[28]. Aligning captures then reasoning about presence versus absence is change detection [29], [30] with classical planar registration [31]–[33] and ECC fallback [34]; our point is that alignment and matching targets should be *disjoint*, with the homography driven by the wall because the crack is what is to be identified.

III. DATA ACQUISITION

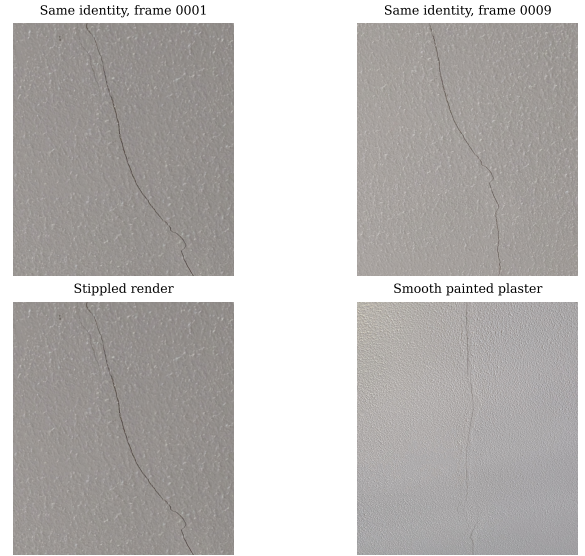
A. Acquisition and composition

The CRACKID frozen benchmark snapshot consists of 140 photographs of 17 exterior walls which were taken at a single location over four half-days using two consumer handheld devices (the data can be obtained through Mendeley Data [35]). The walls have both stippled render (textured) and smooth painted plaster; in the case of the smooth plaster, the only visible features are cracks. The dataset is split into six validation walls (walls 01 to 06; 66 photos, 66 identities) and eleven test walls (walls 07 to 17; 74 photos, 146 identities). Only those identities that appear in at least two photographs are considered to be answerable queries (18 in the validation set and 19 in the test set; $n_Q=305$ in the test set and 439 in the validation set). Additionally, two further walls (walls 18 and 19) have been included and are therefore omitted from all the figures in this paper. Each wall is recorded by a continuous walk-around, with a median time interval of 3 seconds between consecutive frames and a maximum duration of 81 seconds; it is for this reason that the data includes within-visit viewpoint and scale correspondence, which is regarded as an upper bound on multi-visit performance.

Adjacent-frame retrieval. In the case of the 305 test queries which did have an answer, we use the adjacent frame as the starting point for the retrieval process. As the gaps shown in Section VII (ranging from 305 down to 205, then to 149 and finally to 105 for the gaps numbered 0 to 3) make clear, Rank-1 is still achieved as the number of items decreases. This result shows that the method is robust in the face of short-baseline variations rather than indicating a lack of relevance.

B. Annotation, and why it needs auditing

The ORB homography method operates one-to-one, so for each crack, a single click passes from one frame to the next, and the fragment then combines the segments that constitute the individual physical crack. The greatest risk of an incorrect merge is at this stage, such as changing a hard negative into a free positive. Although the number of merges is provided, their accuracy is not. More precisely, 95 of the 212 identities (45



The images shown here are real CrackID images; the top one shows a single identity spanning non-adjacent frames, while the bottom one shows the stippled render alongside the smooth plaster. The labels were obtained via homography propagation and have not been independently audited.

IV. BENCHMARK PROTOCOL

The benchmark measures the visual association component: given a new crack observation, decide whether it matches a recorded defect or has no match in the reference set. Annotated identities serve as proxies for persistent defect records; we measure association accuracy, not duplicate-ticket reduction or full workflow performance.

A. Task formulation and candidate records

A *wall* is one physical wall; an *identity* links observations attributed to the same crack (provenance in Sec. III-B). A *query* is one crack instance; its *gallery* holds eligible instances from other photographs of the same wall, assuming the wall is known. Cross-wall search is out of scope. A query is *known* if its gallery contains the same identity, else *unknown*. Each method scores query-gallery pairs for ranking, thresholded decisions, and identification with rejection. A *query-gallery pair* is the pairwise unit; an *image pair* is the registration/assignment unit. The Hungarian one-to-one constraint applies per image pair only; several observations may share one defect record.

Test counts and prevalence are summarised in Table I for the frozen benchmark snapshot (6 validation walls wall01–wall06; 11 test walls wall07–wall17; wall18–wall19 added later are excluded from all numbers). Registration is attempted for all 301 same-wall test image pairs; 255 yield scored pairs with validity-mask cells (see Sec. VII-B).

B. Unchanged damage under different acquisition conditions

The primary evaluation uses real single-visit walk-arounds per wall, where morphology is treated as approximately unchanged and variation comes from viewpoint, distance, scale,

TABLE I
COUNTING UNITS FOR EVERY METRIC (FROZEN SNAPSHOT, $n_Q=280$ TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	280
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

illumination, and visibility, with same-wall distractors and unknown queries included. Walls are disjoint across validation/test. All methods are used as-is without CRACKID fine-tuning on shared masks (Sec. V); thresholds are calibrated on validation and applied to test. This establishes within-visit correspondence only, not cross-visit or progression robustness.

C. Controlled appearance change and damage progression

A complementary synthetic revisit isolates photometric change from structural growth: ten source photographs from five walls, one seed, same modest affine warp in both conditions. Condition B corrupts appearance (blur, noise, contrast/brightness) with structure intact; condition A extends one tip of the largest component. Full parameters are in code. This compares OSNet with context against HYBRID-SKEL (geometric alignment plus skeleton fallback/blend, $\tau=8.0$ px), which is distinct from the primary REG+CHAMFER reference. With one seed and five walls, results are preliminary checkpoints, not longitudinal evidence.

D. Decision rules and handling of unavailable scores

Pairwise classification accepts a pair at threshold; query-level identification accepts the top gallery instance only at threshold, else rejects; assignment applies the Hungarian maximisation per image block then rejects below the validation threshold. Unscorable pairs are retained: predicted negative for pairwise/assignment F_1 , rank-last for retrieval. A miss on a known query and a correct rejection on an unknown query are thus distinguished from a supported non-match: registration failure is unresolved and may need review, not proof of a new defect. For assignment $F_1=2TP/(2TP+FP+FN)$, correct accepted assignments are true positives, incorrect accepted ones false positives, unmatched known instances false negatives.

E. Evaluation metrics and reference baselines

Retrieval over known queries uses Rank-1, Rank-5, mAP. Pairwise decisions use precision/recall/ F_1 , distinguishing oracle F_1^* from validation-calibrated F_1^{val} . Assignment uses F_1^a . Identification with rejection uses DIR (known accepted-correct) and FAR (unknown accepted); DIR@FAR= 0.1 is the constrained operating point, distinct from the fixed-threshold result. Coverage is the scored-pair fraction and is reported alongside accuracy. References are random ranking and the

best-constant pairwise F_1 of $2p/(1+p)$ (0.305 at test prevalence 0.180; 0.544 on the 0.48 geometric subset; 0.452 at validation prevalence 0.292); full chance levels in Table II. Resampling is by wall given shared capture conditions; with 11 test walls and 19 matchable identities we report point estimates only, with wall-level intervals and score-matrix release as future work (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline tests whether crack location on the wall supports association: it aligns wall surfaces to a common frame and compares projected crack masks. It assumes an approximately planar wall and uses no CRACKID training. Shared UNet++ masks (trained on public crack data only; no CRACKID wall in training) with 200 px / 5 px instance extraction keep inputs consistent across methods (full segmenter settings in code). Masks are treated as given; their error is not quantified.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and stage, not guaranteed per pair. Full detector settings (SIFT with CLAHE contrast enhancement [36]/ratio/robust estimation) are in code.

B. Registration cascade and validity checks

Textured render usually supplies background features; smooth plaster may not. A three-stage cascade is used: masked SIFT, unmasked-SIFT fallback if too few keypoints remain (flagged per pair since the crack then aids alignment), and ECC photometric fallback. Candidates pass plausibility checks (no mirrored/folded quads, scale within $[1/16, 16]$); the surviving stage is recorded per pair and its inlier RMSE sets the comparison tolerance. Unscored pairs are unresolved lack of evidence, not proof of different defects.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance, which tolerates small misalignments that would collapse IoU on narrow cracks. Let $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

with early rejection of distant candidates. Chamfer coverage (fraction within tolerance τ from per-pair RMSE, clamped $[2, 16]$ px, default 6 px; HYBRID-SKEL fixes 8.0 px) is evaluated separately as the scale-free counterpart (Sec. VII-C).

D. Correspondence decisions and limits of spatial evidence

Hungarian one-to-one assignment [37] per image pair plus validation threshold (evaluation, identical for all methods); unscored pairs receive $-\infty$ (predicted negative / rank-last per Sec. IV). An aligned projection into visibly empty wall is a **licensed negative**—positive evidence of absence, stronger than low embedding similarity—while missed detections, out-of-frame regions, and registration failures stay unresolved. REG+CHAMFER targets unchanged damage; progression uses HYBRID-SKEL. Performance is conditional on surface-dependent failures and homography-based annotation provenance.

VI. THE APPEARANCE CEILING

A. Setup

The ten methods are assessed using crack-centred crops and a common validity mask: SIFT and ORB together with RANSAC inliers, SuperGlue and LoFTR using pretrained indoor settings, five embedding-based methods, and re-ID OSNet (using cosine similarity, with all being zero-shot on CRACKID); YOLOv8 makes use of the layer-9 `yolo_v8n.pt` features. The full input sizes and checkpoints are given in the code repository. For the context controls, OSNet and CLIP are used with a fixed backbone: `@ctx1` increases the size of the instance box by $1\times$ in each direction, and `@ctx1@nocrack` fills in the crack in the same view in order to isolate the wall context. The REG+CHAMFER method aligns the walls and then compares the projected masks (as described in Sec. V); the COVERAGE-ONLY control assigns a constant score to the registered pairs without examining the cracks. In the synthetic protocol, OSNet with context is compared to the different HYBRID-SKEL configuration. LightGlue and NetVLAD are regarded as future retrieval baselines.

B. Result: appearance buys almost nothing on the decision

The primary benchmark results, as shown in Table II (11 walls; frozen snapshot; $n_Q=305$ answerable test queries), show that the Rank-1 performance varies between 0.370 and 0.613 (ViT-B/16 obtains the highest score; OSNet achieves the best mAP at 0.497; SuperGlue gives the best crop $\text{DIR@FAR}=0.1$ at 0.384). Yet the oracle F_1^* is in the range from 0.337 to 0.374 and the validation-calibrated F_1^{cv} lies between 0.331 and 0.338, which shows that good ranking performance does not imply accurate decision-making. The small range of the validation-calibrated F_1 indicates difficulties in transferring a decision threshold across the walls. With 19 matchable test identities, no fine-grained ranking among the crop rows is claimed. All the values reported are point estimates.

C. Is it the crack, or the wall behind it?

Contextual information has a positive effect on both ranking and open-set performance, yet even when it is removed the majority of this benefit is still retained: OSNet’s Rank-1 score rises from 0.654 to 0.775 (it reaches 0.711 when the context is removed), DIR@FAR increases from 0.168 to 0.404 (it

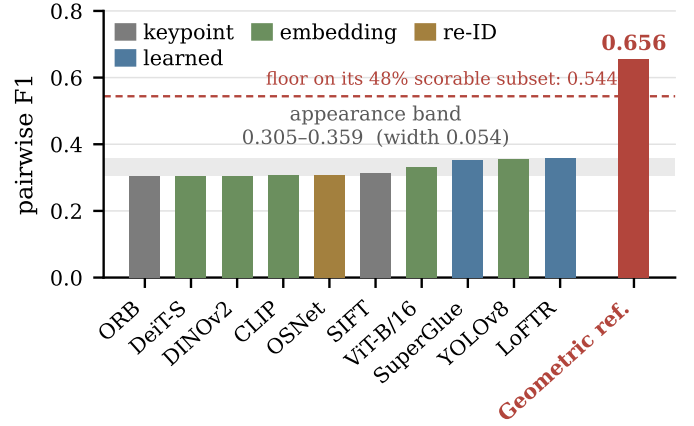


Fig. 1. The appearance ceiling: pairwise F_1^* stays narrow while ranking spreads. Geometry (red) leaves the band but is in the 0.48 pool (own floor 0.544). Table II takes precedence.

is 0.393 when the context is removed), and CLIP’s open-set performance improves from 0.143 to 0.311 (it amounts to 0.286 when the context is removed). The two views that have the context removed outperform all the methods which use tight crops, indicating that the wall location is the main clue. The pairwise F_1^* remains within a narrow range (from 0.307 to 0.361) for the base, context, and erased views, while open-set performance improves by +0.236 for OSNet. Nevertheless, although the methods examined are capable of retrieving within-visit observations, reliable automatic identity decisions are still not possible. A ranked list may be of some help to an operator, but acceptance will need more evidence.

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometric Rank-1 reaches 0.952 vs crop best 0.654 (ViT-B/16 and OSNet tied) and mAP 0.886 vs 0.520 (OSNet) (*scored* 0.48 vs 1.00). Conditioning on coverage as excess over COVERAGE-ONLY, ranking gains stay large, but the open-set number is what coverage cannot explain: $\text{DIR@FAR}=0.1$ of 0.768 vs 0.429 (SuperGlue), 0.404 (best context OSNet+ctx1), 0.038 chance and 0.000 for COVERAGE-ONLY. Crop similarities on self-similar hairlines rarely support rejection, while an aligned empty projection gives a licensed negative. Compute does not explain the gap (LoFTR 8.64s vs OSNet 0.05s per query; Reg 27.9s per query, 26.0s per image pair; timings relative only). RQ2 (geometric part): alignment adds identification beyond coverage alone, conditional on success, surface, and provenance. All numbers are point estimates on 19 matchable identities.

B. Partial coverage, gap structure, and failures

Crop methods score 100% of pairs, geometry 48%; unscored pairs are charged negative/rank-last. Censoring penalises ranking but subsidises pairwise F_1 : the scorable subset keeps 100% of positives but only 36.8% of negatives

TABLE II

MAIN RESULT ON CRACKID TEST (11 WALLS; FROZEN SNAPSHOT, $n_Q=305$). F_1^* IS COMPUTED USING AN ORACLE THRESHOLD ON THE TEST SET; ΔF_1^* REPRESENTS THE EXCESS OVER THE FULL-POOL FLOOR. $F_1^{\text{@v}}$ IS CALCULATED USING THE VALIDATION-CALIBRATED THRESHOLD. BOLD FONT INDICATES THE BEST CROP ROW IN EACH COLUMN. THE GEOMETRIC ROW IS BASED ON A SUBSET OF SIZE 0.48 WITH A FLOOR OF 0.544. THEREFORE, ITS ΔF_1^* IS NOT DIRECTLY COMPARABLE TO THAT OF THE CROP ROWS; SEE SEC. VII-B. FOR DETAILS ON SCORING, CHANCE LEVELS, AND COMPUTATIONAL COSTS, REFER TO SEC. IV THROUGH SEC. VII-B. THE METRIC s/query IS DEFINED AS $\text{TOTAL_S}/n_Q$ ON A SINGLE WORKSTATION (RELATIVE TIMINGS ONLY).

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{\text{@v}}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [22]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

(prevalence 0.180→0.373, floor 0.305→0.544). COVERAGE-ONLY—constant on registered pairs without inspecting cracks—thus scores 0.544 pairwise F_1 , 0.533 mAP and 0.515 Rank-1: much of the ranking/pairwise margin is *which pairs align*. The open-set lead (0.768 vs 0.000) is the exception we lead with. Failures are wall-dependent (two test walls 0%; full per-wall table in supplement), so aggregates are not uniform across surfaces. All 280 answerable queries are answered by the adjacent frame, so headline ranking is short-baseline correspondence; gap sweeps (280→105 at gap 3) leave Rank-1 flat on a shrinking pool rather than proving irrelevance. Registration follows image sharpness over frame gap (wall-clustered Wald $z=5.11$, $p=3.3 \times 10^{-7}$ vs -2.78 , $p=5.4 \times 10^{-3}$; details in supplement); one sharpness gate (≥ 10) raises registration to 56.1% on the filtered set at the cost of one wall, suggesting capture-quality guidance, not full-population coverage.

C. Ablations and cost

Coverage (scale-free) costs 0.240 mAP and 0.103 Rank-1 versus Chamfer distance, with DIR nearly unchanged (+0.014): the comparison rule moves ranking, not decisions. Disabling crack exclusion leaves ranking/pairwise unchanged but raises DIR by +0.057; masking is retained for evidence independence, not observed accuracy here. ECC adds $\sim +0.03$ Rank-1 / +0.02 mAP with DIR unchanged. Cost is 26.0 s per image pair (7817.3 s over 301 test pairs, i.e. 27.9 s per query; Chamfer-only thereafter), about $3 \times$ LoFTR per query and more than two orders above the cheapest embedding; absolute timings are workstation-dependent and SuperGlue/LoFTR were timed in a separate run.

Protocol	Method	n_q	R@1	mAP	DIR@FAR _{0.1}	$F_1^{\text{@v}}$	F_1^a
A (growth)	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	OSNet+ctx	69	0.826	0.647	0.739	0.601	0.631
B (appear.)	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	OSNet+ctx	82	0.671	0.570	0.663	0.594	0.609

TABLE III

HYBRID-SKEL VS OSNET+CTX ON SYNTHETIC REVISIT; F_1 AT VALIDATION THRESHOLD. ONE SEED, FIVE WALLS, NO RESAMPLING: CHECKPOINT, NOT STABILITY CLAIM. HYBRID-SKEL $\neq$ TABLE II REG+CHAMFER.

D. A synthetic revisit protocol

Ten source photographs from five walls (one seed) are warped identically with (A) single-tip growth or (B) photo-metric corruption, ranked against same-wall real instances (full recipe in code; warp modest vs real p90 tilt 75°). HYBRID-SKEL (alignment plus skeleton fallback/blend) differs from Table II REG+CHAMFER, so rows test controlled change, not the main configuration.

OSNet+ctx dominates ranking/open-set (0.826/0.671 and 0.739/0.663 vs 0.275/0.274 and 0.343/0.342), while HYBRID-SKEL holds assignment F_1^a (0.646/0.640 vs 0.631/0.609) with pairwise $F_1^{\text{@v}}$ close (0.569/0.574 vs 0.601/0.594); no assignment cell came from pure homography alone ($H\%=0.0$). Ordering thus reverses with metric (RQ3): the Table II advantage does not imply multi-visit robustness, and single-seed controlled growth does not establish real progression performance.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This work establishes a first measurable formulation of within-visit crack re-identification and a geometric reference

baseline. The results are point estimates on a small, wall-clustered sample with single-annotator labels and partial registration coverage, so each finding below is paired with the concrete step that would confirm or extend it.

- 1) **Scale to multi-wall, multi-visit data.** Collect more walls, surface families, and annotators to support wall-level confidence intervals and reliable method ranking. Most importantly, add a second capture session at deliberately varied distances and lighting to convert the within-visit measurement into a genuine cross-visit benchmark for ticket management backends.
- 2) **Add provenance-free annotation.** All clicks were human-validated by a single annotator with full triage of unclaimed points, but agreement is unmeasured and the geometric advantage remains provenance-conditioned. A small subset labeled entirely without homography propagation, plus released per-pair score matrices for sensitivity analysis, wall-clustered intervals, and threshold-free AP/AUROC, would quantify how much of the open-set gap is independent of annotation provenance.
- 3) **Close the registration coverage gap.** The baseline answers 48% of pairs (97/301 image pairs; 15 328/31 822 valid cells) and scores zero on smooth-plaster walls. Surface-adaptive registration — learned local features, direct photometric alignment, or sparse depth from consumer devices — would push coverage toward full-set evaluation and further decouple the geometry pipeline from homography-based provenance.
- 4) **Learn from geometry-derived supervision.** The baseline is a non-learned reference, not a deployment target. Using the registration pipeline as a teacher for pseudo-labels, or fine-tuning a foundation-model backbone on homography-derived correspondence, would retain the location-based advantage on pairs registration cannot answer.
- 5) **Evaluate the ticket lifecycle.** Define task-level metrics — duplicate rate and missed-new-defect rate — and evaluate segmentation, re-identification, and ticket assignment jointly in a closed-loop simulation of real inspection workflows.

Together, these steps turn the present within-visit reference into a deployable multi-visit re-identification component for operational ticket management.

IX. CONCLUSION

We have developed a dataset that enables the testing of crack re-identification: CrackID, which includes 140 photographs taken from 17 walls and covering 212 identities in the frozen snapshot (6 of these being validation walls and 11 being test walls; walls 18 and 19 were excluded); the source of the annotations (45

The key result of the benchmark is negative and structural in nature: when a decision is required, ten of the zero-shot matchers from four different families achieve a score between +0.000 and +0.054 above the 0.305 best-constant floor (F_1^* ranging from 0.305 to 0.359), and eight out of

the ten entries are at or below the floor when assessed at the validation threshold (F_1^{val} ranging from 0.293 to 0.351; LoFTR alone scoring +0.046). Determining a person’s identity from a gallery is a separate ability to that of making a decision. The problem does not lie merely in the appearance of cracks: a geometric reference baseline that uses wall texture as a reference while the cracks are masked and Chamfer-matches are projected achieves a DIR@FAR=0.1 of 0.768 compared to 0.429 for the best crop method and 0.404 for the best context method, correctly identifying 48% of the pairs, and a COVERAGE-ONLY control that makes use of that alignment without looking at the crack itself obtains a score of 0.000 on the open-set metric (although it has a pairwise F_1 of 0.544 on its own 0.544 floor).

As far as operation is concerned, the advantage in the open-set scenario results from an aligned pair serving as a *licensed negative*, that is, positive evidence of absence, since a crop embedding can only show low similarity; it is still not clear whether this benefit is independent of homography-based annotation and it must therefore be regarded as depending on the source. Two issues remain to be addressed: carrying out a real second capture session changes the benchmark from one based on viewpoint to one based on cross-visit, and having a second annotator together with a homography-free audit subset would turn the single-annotator triage into a measured agreement rate. The 48

ACKNOWLEDGMENT

This work was supported by “R2UTechnologies — modular systems” (C644876810-00000019), by “R2UTechnologies” Consortium, cofinanced by NextGenerationEU, through the Incentive System “Agendas para a Inovação Empresarial” (“Agendas for Business Innovation”), within the Recovery and Resilience Plan (PRR).

REFERENCES

- [1] S. Kulkarni, S. Singh, D. Balakrishnan, S. Sharma, S. Devunuri, and S. C. R. Korlapati, “CrackSeg9k: A collection and benchmark for crack segmentation datasets and frameworks,” 2022.
- [2] C. Benz and V. Rodehorst, “OmniCrack30k: A benchmark for crack segmentation and the reasonable effectiveness of transfer learning,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2024.
- [3] K. Zhou, Y. Yang, A. Cavallaro, and T. Xiang, “Omni-scale feature learning for person re-identification,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 3702–3712.
- [4] Y. Qian, J. Barthelemy, E. Karuppiyah *et al.*, “Identifying re-identification challenges: Past, current and future trends,” *SN Computer Science*, vol. 5, p. 937, 2024.
- [5] Q. Zou, Y. Cao, Q. Li, Q. Mao, and S. Wang, “CrackTree: Automatic crack detection from pavement images,” *Pattern Recognition Letters*, vol. 33, no. 3, pp. 227–238, 2012.
- [6] Y. Liu, J. Yao, X. Lu, R. Xie, and L. Li, “DeepCrack: A deep hierarchical feature learning architecture for crack segmentation,” *Neurocomputing*, vol. 338, pp. 139–153, 2019.
- [7] M. Eisenbach, R. Stricker, D. Seichter, K. Amende, K. Debes, M. Sessmann, D. Ebersbach, U. Stoeckert, and H.-M. Gross, “How to get pavement distress detection ready for deep learning? a systematic approach,” in *Int. Joint Conf. Neural Networks (IJCNN)*, 2017, pp. 2039–2047.

- [8] C. Benz and V. Rodehorst, "CrackStructures and CrackEnsembles: The power of multi-view for 2.5d crack detection," in *Proceedings of the Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 5990–5999.
- [9] L. Parente, E. Falvo, C. Castagnetti, F. Grassi, F. Mancini, P. Rossi, and A. Capra, "Image-based monitoring of cracks: Effectiveness analysis of an open-source machine learning-assisted procedure," *Journal of Imaging*, vol. 8, no. 2, p. 22, 2022.
- [10] S. Harb, P. Achancaray, M. Maboudi, and M. Gerke, "Multi-temporal crack segmentation in concrete structures using deep learning approaches," *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. X-G-2025, pp. 341–348, 2025.
- [11] X. Sun and P. Chang, "Robust perspective correction for real-world crack evolution tracking in image-based structural health monitoring," 2025.
- [12] NEC Corporation, "Photographing guide device," US Patent 12,223,639 B2 (priority 2019), 2025.
- [13] S. Liao, Z. Mo, Y. Hu, and S. Z. Li, "Open-set person re-identification," arXiv preprint arXiv:1408.0872, 2014.
- [14] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *International Journal of Computer Vision*, vol. 60, no. 2, pp. 91–110, 2004.
- [15] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *IEEE International Conference on Computer Vision (ICCV)*, 2011, pp. 2564–2571.
- [16] P.-E. Sarlin, D. DeTone, T. Malisiewicz, and A. Rabinovich, "SuperGlue: Learning feature matching with graph neural networks," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4938–4947.
- [17] J. Sun, Z. Shen, Y. Wang, H. Bao, and X. Zhou, "LoFTR: Detector-free local feature matching with transformers," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 8922–8931.
- [18] A. Dosovitskiy, L. Beyer, A. Kolesnikov *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021.
- [19] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers and distillation through attention," in *International Conference on Machine Learning (ICML)*, 2021, pp. 10 347–10 357.
- [20] A. Radford, J. W. Kim, C. Hallacy *et al.*, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [21] M. Oquab, T. Darcet, T. Moutakanni *et al.*, "DINOv2: Learning robust visual features without supervision," *Transactions on Machine Learning Research*, 2024.
- [22] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," <https://github.com/ultralytics/ultralytics>, 2023.
- [23] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, "LightGlue: Local feature matching at light speed," in *IEEE/CVF Int. Conference on Computer Vision (ICCV)*, 2023, pp. 17 627–17 638.
- [24] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "NetVLAD: CNN architecture for weakly supervised place recognition," in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 5297–5307.
- [25] L. Maiano, A. Montuschi, M. Caserio *et al.*, "A deep-learning-based antifraud system for car-insurance claims," *Expert Systems with Applications*, vol. 231, p. 120644, 2023.
- [26] M. J. Hasan, C. K. Nguyen, Y. L. Boo, H. Jahani, and K.-L. Ong, "Vehicle damage detection using artificial intelligence: A systematic literature review," *WIREs Data Mining and Knowledge Discovery*, vol. 15, p. e70027, 2025.
- [27] S. A. Pérez-Zarate, D. Corzo-García, J. L. Pro-Martín *et al.*, "Automated car damage assessment using computer vision: Insurance company use case," *Applied Sciences*, vol. 14, no. 20, p. 9560, 2024.
- [28] D. Mallios, L. Xiaofei, N. McLaughlin, J. M. Del Rincon, C. Galbraith, and R. Garland, "Vehicle damage severity estimation for insurance operations using in-the-wild mobile images," *IEEE Access*, vol. 11, pp. 78 644–78 655, 2023.
- [29] K. Sakurada and T. Okatani, "Change detection from a street image pair using CNN features and superpixel segmentation," in *British Machine Vision Conference (BMVC)*, 2015.
- [30] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Autonomous Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.
- [31] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge University Press, 2004.
- [32] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," in *Readings in Computer Vision*. Morgan Kaufmann, 1987, pp. 726–740.
- [33] D. Barath, J. Matas, and J. Noskova, "MAGSAC: Marginalizing sample consensus," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 10 197–10 205.
- [34] G. D. Evangelidis and E. Z. Psarakis, "Parametric image alignment using enhanced correlation coefficient maximization," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 30, no. 10, pp. 1858–1865, 2008.
- [35] D. Ferramenta, "CrackID," Mendeley Data, Version 1, <https://doi.org/10.17632/t749fc4x44.1>, 2026.
- [36] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*. Academic Press, 1994, pp. 474–485.
- [37] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, no. 1-2, pp. 83–97, 1955.